

IT3051 | FUNDAMENTALS OF DATA MINING | MINI PROJECT 2026

Obesity Risk Intelligence System

Complete project plan, milestone guide, and equal four-member work division

4

equal full-project contributors

7

obesity-risk target classes

12

assignment workflow stages

Repository: Obesity-Risk-Intelligence-System | **Integration branch:** main

Educational use only. Predictions are not medical diagnoses or treatment recommendations.

1. Project Direction

The team will build an educational machine-learning system that classifies a person's obesity-risk category using demographic, physical, nutritional, and lifestyle information. The task is a seven-class classification problem with **NObesdad** as the target variable.

Important approval gate: The dataset must be validated by the instructor before EDA, preprocessing, model training, or application development. Until approval is recorded, the repository contains planning and structure only.

Users and stakeholders

- Students and academic evaluators
- Researchers and data-mining learners
- Demonstration users and non-technical stakeholders

System outcome

- Validated user inputs
- Predicted obesity-risk category
- Clear confidence/probability information
- Educational explanation and limitations

2. Dataset Proposal

Item	Project decision
Dataset	Kaggle Playground Series Season 4 Episode 2 obesity-risk dataset
Source	kaggle.com/competitions/playground-series-s4e2/data
Task and target	Seven-class classification; target: NObesdad
Expected size	20,758 training records and 18 columns; verify from the downloaded source
Predictive features	8 numerical and 8 categorical features; id is an identifier and is normally excluded
Suitability	Combines physical, demographic, nutritional, and lifestyle variables with a clear categorical target, supporting EDA, preprocessing, model comparison, tuning, and system integration
Limitations	Public competition data may contain synthetic or simulated records. Results must not be presented as population-level clinical conclusions.

Expected numerical features: Age, Height, Weight, FCVC, NCP, CH20, FAF, TUE

Expected categorical features: Gender, family_history_with_overweight, FAVC, CAEC, SMOKE, SCC, CALC, MTRANS

Record the dataset version, download date, source, citation, license, and instructor validation evidence. Never commit Kaggle credentials or private health information.

3. Repository and Branch Plan

```
Obesity-Risk-Intelligence-System/
├─ backend/      prediction API and service
├─ data/         raw, interim, and processed data
├─ docs/         plans, decisions, evidence, and report material
├─ frontend/     user-facing prediction interface
├─ models/       trained model and metadata
├─ notebooks/    ordered analysis notebooks
├─ reports/      figures, results, and report assets
├─ src/          reusable data and ML code
├─ tests/        unit and integration tests
├─ README.md
└─ LICENSE
```

Equal responsibility rule: Branch names identify members, not isolated project parts. Every member contributes to the full project, reviews other work, participates in testing, writes report/presentation material, and can explain the complete system during individual evaluation.

Branch	Member	Lead coordination area	Full-project expectation
member-1	Member 1	Problem definition, dataset proposal, and model comparison	Contribute to and review all stages
member-2	Member 2	Data understanding, EDA, and model tuning	Contribute to and review all stages
member-3	Member 3	Preprocessing, feature engineering, and backend	Contribute to and review all stages
member-4	Member 4	Integration, evidence, frontend, and testing	Contribute to and review all stages

Branch rules

1. Start each branch from the latest `main`.
2. Use clear commits and open pull requests into `main`.
3. Obtain at least one review before merging.
4. Record the lead, reviewers, files changed, evidence, and personal contribution for each stage.
5. Do not merge work that the author cannot explain.

4. Full Project Workflow

STAGE 1	Problem understanding Define the real-world scenario, prediction objective, target variable, users, stakeholders, decisions, inputs, outputs, and educational/non-clinical scope.
STAGE 2	Dataset identification and validation Prepare the proposal with source, citation, context, dimensions, feature groups, task, suitability, data-quality observations, limitations, ethics, privacy, licensing, and accessibility. Obtain approval before development.
STAGE 3	Data understanding and EDA Inspect structure, meanings, types, distributions, relationships, missing values, duplicates, identifiers, invalid values, outliers, unusual observations, class balance, visualisations, and leakage risks.
STAGE 4	Preprocessing and feature engineering Handle missing, duplicate, invalid, and justified outlier cases; encode categorical variables; scale where required; decide identifier handling; select or engineer features; split data without leakage.
STAGE 5	Progress Evaluation 1 Individual viva on the problem, dataset, EDA, quality issues, preprocessing, feature engineering, leakage prevention, and personal contribution.
STAGE 6	Model development Implement and compare at least four suitable algorithms using an appropriate validation strategy and classification metrics. Record experiments and explain why results differ.
STAGE 7	Model optimization and selection Tune suitable models, investigate feature/model improvements, compare tuned and baseline results, and justify the final model.
STAGE 8	Progress Evaluation 2 Individual viva on algorithms, validation, metrics, tuning, optimization, interpretation, final model selection, and personal contribution.

STAGE 9	Backend development Load the final model and preprocessing pipeline, validate inputs, generate predictions, return meaningful results, and handle invalid or missing inputs.
STAGE 10	Frontend and user experience Provide clear input controls, validation, prediction results, understandable explanations, and a working frontend-backend end-to-end flow.
STAGE 11	Technical report Cover problem, requirements, dataset, EDA, preprocessing, features, algorithms, evaluation, tuning, final model, architecture, implementation, testing, limitations, future work, and contributions.
STAGE 12	Final presentation and demonstration Explain the business problem and evidence-based solution to non-technical stakeholders, demonstrate the system, communicate recommendations, and answer questions individually.

5. Progress Evaluation 1

Assessment

Weight: 30%

Type: Individual viva

Coverage: Stages 1-4 and personal contribution

Dates in guideline

23, 26, or 27 September 2026

KandyUni and Northern Uni dates will be communicated by staff.

Every member must prepare

- Problem scenario, prediction objective, target, task type, users, and requirements
- Dataset name, source, citation, context, records, features, target, suitability, and limitations
- Instructor validation status and evidence
- Data types, missing values, duplicates, invalid values, outliers, and class balance
- Important EDA observations and visualisations
- Preprocessing, feature-engineering, split, and leakage-prevention decisions
- Personal contribution, branch commits, reviews, and decisions

Gate: Do not present unverified assumptions as findings. Mark them as pending until confirmed against the approved dataset.

6. Progress Evaluation 2

Assessment

Weight: 30%

Type: Individual viva

Coverage: Stages 6-7 and personal contribution

Dates in guideline

30 September, 3, or 4 October 2026

KandyUni and Northern Uni dates will be communicated by staff.

Every member must prepare

- Reasons for selecting at least four algorithms
- Validation strategy and suitability of metrics
- Baseline and tuned model comparison
- Hyperparameter search method and optimization decisions
- Interpretation of model strengths, weaknesses, errors, and limitations
- Final model selection and evidence-based justification
- Personal contribution and ability to explain the complete modelling workflow

7. Final Deliverables and Dates

Deliverable	Required content	Guideline date
Approved dataset and evidence	Proposal, source, citation, validation record, and version details	Validation deadline: 20 September 2026
Source code	EDA, preprocessing, modelling, backend, frontend, tests, and integration code	Before final submission
Trained final model	Final model with the correct preprocessing pipeline and metadata	Before final submission
Technical report	Complete technical workflow, results, architecture, testing, limitations, and contributions	6 October 2026
Presentation and system demo	Non-technical explanation, evidence, recommendations, working application, and individual answers	7, 10, or 11 October 2026

Dates for KandyUni and Northern Uni students will be provided by staff members where stated in the assignment document.

8. Four-Member Work Division

Member	Lead first draft	Must also contribute to	Evidence to maintain
Member 1 member-1	Problem, dataset proposal, source/citation, model comparison	EDA, preprocessing, tuning, backend, frontend, tests, report, presentation	Proposal, model comparison records, review comments, personal log
Member 2 member-2	Data dictionary, EDA, distributions, tuning	Dataset validation, preprocessing, model experiments, backend, frontend, tests, report, presentation	EDA notebook, findings, tuning log, review comments, personal log
Member 3 member-3	Preprocessing, feature engineering, backend	Problem, EDA, models, tuning, frontend, testing, report, presentation	Preprocessing plan, pipeline decisions, API tests, review comments, personal log
Member 4 member-4	Integration, evidence, frontend, testing	Problem, dataset, EDA, preprocessing, models, tuning, backend, report, presentation	Requirements matrix, integration tests, UI evidence, review comments, personal log

Equal contribution standard: Lead work is coordination, not exclusive ownership. Every member must be able to reproduce, review, explain, and defend the full project.

9. Final Quality Checklist

- ☐ Instructor validation is recorded before development proceeded
- ☐ Four suitable algorithms were implemented and compared
- ☐ Metrics and validation match the multi-class classification task

- ❑ Final model uses the same preprocessing pipeline used during training
 - ❑ Backend validates inputs and returns clear predictions
 - ❑ Frontend and backend work together end to end
 - ❑ Testing evidence covers valid, invalid, missing, and representative inputs
 - ❑ Report, slides, source, model, and evidence are complete
 - ❑ Each member's contribution is documented and individually understood
-

Source: IT3051 Fundamentals of Data Mining Mini Project Assignment 2026 and the proposed Kaggle dataset page. Verify the live Kaggle citation metadata and any staff-issued date changes before submission.